

On the order of Erdős-Rogers functions

February 8, 2024

Abstract

For an integer $n \geq 1$, the Erdős-Rogers function $f_s(n)$ is the maximum integer m such that every n -vertex K_{s+1} -free graph has a K_s -free subgraph with m vertices. It is known that for all $s \geq 3$, $f_s(n) = \Omega(\sqrt{n \log n} / \log \log n)$ as $n \rightarrow \infty$. In this paper, we show that for all $s \geq 3$,

$$f_s(n) = O(\sqrt{n} \log n).$$

This improves previous bounds of order $\sqrt{n}(\log n)^{2(s+1)^2}$ by Dudek, Retter and Rödl.

1 Introduction

For an integer $s \geq 2$, the *Erdős-Rogers function* $f_s(n)$ is the maximum integer m such that every n -vertex K_{s+1} -free graph has a K_s -free subgraph with m vertices. The determination of $f_2(n)$ is almost equivalent to determining the triangle-complete graph Ramsey numbers, since $r(3, f_2(n)) \leq n < r(3, f_2(n) + 1)$. These quantities $r(3, t)$ are known to within a constant factor [1, 4, 7, 11, 18], and from this one deduces $f_2(n)$ has order of magnitude $\sqrt{n \log n}$ as $n \rightarrow \infty$. As observed by Dudek and the first author, the arguments for lower bounds for $f_2(n)$ generalize to $f_s(n)$ for $s \geq 3$. Shearer [19] showed that any n -vertex K_{s+1} -free graph of maximum degree d has an independent set of size $\Omega((n \log d)/(d \log \log d))$, and the neighborhood of a vertex of degree d is a K_s -free induced subgraph. Therefore for all $s \geq 3$,

$$f_s(n) = \Omega\left(\frac{\sqrt{n \log n}}{\log \log n}\right). \quad (1)$$

Wolfovits [21] proved $f_3(n) = O(\sqrt{n}(\log n)^{120})$, and it was shown by Dudek, Retter, Rödl [6] that $f_3(n) = O(\sqrt{n}(\log n)^{32})$ and more generally that $f_s(n) = O(\sqrt{n}(\log n)^{2(s+1)^2})$. In this short paper, we significantly improve these bounds on the Erdős-Rogers functions $f_s(n)$ as follows:

Theorem 1. *For each fixed $s \geq 3$,*

$$f_s(n) = O(\sqrt{n} \log n). \quad (2)$$

The proof of Theorem 1 involves a combination of the ideas of Wolfovits [21] and Dudek, Retter, Rödl [6] with the construction of Mattheus and the second author [14], but does not make use of the method of containers as in [14] or [10]. We did not expend too much effort in optimizing the implicit constant in the bound on $f_s(n)$ in Theorem 1; from the proof one may obtain $f_s(n) \leq 2^{100s} \sqrt{n} \log n$ for $n \geq 2$, which shows $f_s(n) = n^{1/2+o(1)}$ for $s = o(\log n)$.

Notation. For a graph G , we write $V(G)$ for the vertex set of G and $E(G)$ for the edge set of G . For a set $X \subseteq V(G)$, let $G[X]$ denote the subgraph of G induced by X , namely the graph with vertex set X and edge set $\{e \in E(G) : e \subseteq X\}$.

2 Tools from probability

We refer to the book by Alon and Spencer [2] for a reference on probabilistic methods in combinatorics. The first result we need is the *Chernoff Bound* (see Alon and Spencer [2]) for concentration of binomial random variables.

Proposition 1. (Chernoff Bound) *Let Z be a binomial random variable with mean μ . Then for any real $\epsilon \in [0, 1]$,*

$$\begin{aligned} \Pr(Z > (1 + \epsilon)\mu) &\leq \exp\left(-\frac{\epsilon^2\mu}{4}\right) \quad \text{and} \\ \Pr(Z < (1 - \epsilon)\mu) &\leq \exp\left(-\frac{\epsilon^2\mu}{2}\right) \end{aligned} \tag{3}$$

The next proposition is derived from *Janson's inequality* [2, 9]. Let χ be a coloring of an n -element set Y with s colors, with color classes $Y_1, Y_2, \dots, Y_s$. Then the *random s -partite graph* $G_{n,\rho}(\chi)$ samples edges independently with probability ρ from the complete multipartite graph with parts $Y_1, Y_2, \dots, Y_s$. The following technical proposition is derived in a standard way from Janson's inequality [2, 9], and we give a proof in the appendix:

Proposition 2. (via Janson's inequality) *Let $s \geq 3$, $n \geq 2^{40s}$ and $\rho = (8s/n)^{2/s}$, and let χ be an s -coloring of an n -element set whose color classes have size at least $n/2s$ each. Then*

$$\Pr(K_s \not\subseteq G_{n,\rho}(\chi)) \leq \exp(-2^{2s-4}n). \tag{4}$$

We shall finally require the *Lovász local lemma* [2, 13] in the following form. We write $\overline{A}$ for the complement of an event A in a probability space.

Proposition 3. (Lovász local lemma) *Let $A_1, A_2, \dots, A_n$ be events in some probability space and suppose that for some set $J_i \subset \{1, 2, \dots, n\}$, A_i is mutually independent of $\{A_j : j \notin J_i \cup \{i\}\}$. If there exist real numbers $\gamma_i \in [0, 1]$ such that*

$$\Pr(A_i) \leq \gamma_i \prod_{j \in J_i} (1 - \gamma_j), \tag{5}$$

then

$$\Pr\left(\bigcap_{i=1}^n \overline{A_i}\right) > 0.$$

3 Hermitian unitals and O’Nan configurations

The proof of Theorem 1 appeals to a construction from projective geometry. We briefly describe the geometry here and the tools from probabilistic combinatorics that we use. Further geometric background is given in Barwick and Ebert [3], Brouwer and van Maldeghem [5] and Piper [17].

3.1 Hermitian unitals in brief

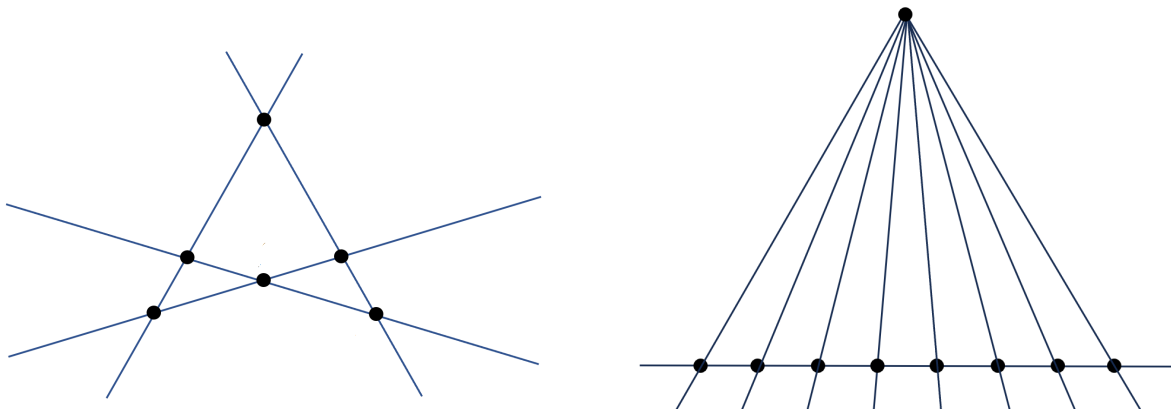
For a partial linear space $\mathcal{H}$, we let $P(\mathcal{H})$ denote the set of points of $\mathcal{H}$ and $L(\mathcal{H})$ denote the set of lines of $\mathcal{H}$. A *unital* in the projective plane $\text{PG}(2, q^2)$ is a set $\mathcal{U}$ of $q^3 + 1$ points such that every line of $\text{PG}(2, q^2)$ intersects $\mathcal{U}$ in 1 or $q + 1$ points – the latter are referred to as *secants*. A *classical* or *Hermitian unital* $\mathcal{H}_q$ is a partial linear space described in homogeneous co-ordinates as the following set of one-dimensional subspaces of $\mathbb{F}_{q^2}^3$:

$$P(\mathcal{H}_q) = \{ \langle x, y, z \rangle \subset \mathbb{F}_{q^2}^3 : x^{q+1} + y^{q+1} + z^{q+1} = 0 \}.$$

Here arithmetic is in the finite field $\mathbb{F}_{q^2}$, and $\langle x, y, z \rangle$ is the one-dimensional subspace of $\mathbb{F}_{q^2}^3$ generated by (x, y, z) . Then $L(\mathcal{H}_q)$ consists of the intersections of secant lines with $P(\mathcal{H}_q)$, so that there are $q^2(q^2 - q + 1)$ lines in $\mathcal{H}_q$, each containing exactly $q + 1$ points of $\mathcal{H}_q$.

3.2 O’Nan configurations and fans

One of the remarkable features of the Hermitian unital is that it does not contain the so-called *O’Nan configuration*, namely the configuration of four lines and six points in the left figure below:



O’Nan configuration and s-fan

Definition 1. (s-fan) For $s \geq 3$, an *s-fan* is a set of s pairwise intersecting lines such that $s - 1$ of the lines are concurrent with a point whereas the remaining line is not concurrent with that point. For $s \geq 4$, the unique point contained in $s - 1$ lines is the point of concurrency of the *s-fan*.

An illustration of an *s-fan* is shown in the right figure above. The fact that $\mathcal{H}_q$ does not contain the figure on the left was first proved by O’Nan [16] (see Mattheus and the second author for a short linear-algebraic proof [14]). The following lemma is a straightforward consequence:

Lemma 1. *If $s \geq 3$ lines in $\mathcal{H}_q$ pairwise cross, then they are concurrent with some point of $\mathcal{H}_q$ or they form an s -fan.*

Proof. Let $K \subseteq L(\mathcal{H}_q)$ be a configuration of s pairwise crossing lines. In the case $s = 3$, if the lines are not all concurrent with some point, then they form a triangle, which is an s -fan. The case $s = 4$ follows from the fact that $\mathcal{H}_q$ contains no O’Nan configuration. If $s > 4$, then at least three of the lines $\ell_1, \ell_2, \ell_3 \in K$ are concurrent with some point $p \in \mathcal{H}_q$. Suppose for a contradiction that two of the lines $\ell_4, \ell_5 \in K$ are not concurrent with p . Then three of $\ell_1, \ell_2, \ell_4, \ell_5$ are concurrent with some $p' \in P(\mathcal{H}_q) \setminus \{p\}$. This implies $\ell_1, \ell_2, \ell_3, \ell_4, \ell_5$ are concurrent with p' , contradicting that ℓ_4, ℓ_5 are concurrent with p . $\square$

4 Random sampling

To prove Theorem 1, we require for $s \geq 3$ an n -vertex K_{s+1} -free graph H such that every induced subgraph of H with substantially more than about $\sqrt{n} \log n$ vertices contains a copy of K_s . The overview of the construction of H is as follows, where asymptotic notation is with respect to a growing prime power q .

First we randomly sample points from the Hermitian unital $\mathcal{H}_q$ with probability roughly $\Theta(\log q / (q + 1))$ for to obtain a partial linear space $\mathcal{H}$. We show in Section 4.1 via the Chernoff bound, Proposition 1, that with positive probability, $|P(\mathcal{H})| = \Theta(q^2 \log q)$, $|L(\mathcal{H})| = q^2(q^2 - q + 1)$, each line in $L(\mathcal{H})$ has size $\Theta(\log q)$, and the number of $(s + 1)$ -fans containing any given pair of intersecting lines in $\mathcal{H}$ is $\Theta(\log q)^s$.

We define the *intersection graph* G whose vertex set is $L(\mathcal{H})$ and whose edges are pairs of intersecting lines of $\mathcal{H}$. The graph G has $q^2(q^2 - q + 1)$ vertices and every edge of G is contained in $\Theta(\log q)^s$ copies of K_{s+1} in G . Most importantly,

$$\begin{aligned} &\text{the vertex set of each } K_{s+1} \subseteq G \text{ is either an } (s + 1)\text{-fan in } \mathcal{H} \text{ or} \\ &\text{is a set of lines of } \mathcal{H} \text{ all concurrent with some point in } \mathcal{H}. \end{aligned} \tag{A}$$

We eliminate the copies of $K_{s+1} \subseteq G$ corresponding to $s + 1$ lines of $\mathcal{H}$ concurrent with a point p of $\mathcal{H}$ by randomly s -coloring the set $\mathcal{H}_p$ of lines concurrent with p , independently over different points of $\mathcal{H}$, and retaining only those edges $\{\ell, \ell'\}$ of G such that ℓ and ℓ' have different colors in the coloring of $\mathcal{H}_p$ where $p = \ell \cap \ell'$. By (A), this graph G_χ has the following property:

$$\text{the vertex set of each } K_{s+1} \subseteq G_\chi \text{ is an } (s + 1)\text{-fan.} \tag{B}$$

We eliminate the copies of K_{s+1} in G which correspond to $(s + 1)$ -fans in $\mathcal{H}$ by randomly sampling edges of G independently with suitable probability $\rho = \Theta(\log q)^{-2/s}$ to obtain a random graph $G_\rho \subseteq G$. Let H be the intersection of G_ρ and G_χ , that is, $V(H) = V(G)$ and $E(H) = E(G_\rho) \cap E(G_\chi)$. We apply the Lovász local lemma (Proposition 3) in Section 5 to show that with positive probability, the graph H is K_{s+1} -free and, for a constant C depending only on s , every set of $Cq^2 \log q$ vertices of H induces a subgraph of H containing K_s . Then $|V(H)| = n = q^2(q^2 - q + 1)$ and so $f_s(n) \leq Cq^2 \log q$. The distribution of primes q allows us to deduce $f_s(n) = O(\sqrt{n} \log n)$ for all n .

4.1 Randomly sampling points

The first step in our construction is to randomly sample points from $\mathcal{H}_q$ to obtain a partial linear space $\mathcal{H}$ with the properties listed below, which essentially uses only the Chernoff Bound, Proposition 1.

Lemma 2. *For any integers $s \geq 3$ and $a \geq 128$ and any prime power $q \geq a \log q$, there exists a partial linear space $\mathcal{H} = \mathcal{H}_{a,q,s}$ such that*

- (i) *The line set L of $\mathcal{H}$ has size $q^2(q^2 - q + 1)$.*
- (ii) *The point set P of $\mathcal{H}$ has size at most $2aq^2 \log q$ and at least $a(q^2 \log q)/2$.*
- (iii) *Each line of $\mathcal{H}$ has at least $(a \log q)/2$ points.*
- (iv) *The number of $(s+1)$ -fans in $\mathcal{H}$ containing a pair of lines in L is at most $k = (2a \log q)^s$.*
- (v) *Every $s+1$ pairwise intersecting lines in $\mathcal{H}$ is concurrent with a point or is an $(s+1)$ -fan.*

Proof. Sample points of $\mathcal{H}_q$ randomly and independently with probability $(a \log q)/(q+1)$ and let $\mathcal{H}$ be the partial linear space whose point set is the set P of sampled points and whose line set is $L = \{\ell \cap P : \ell \in L(\mathcal{H}_q)\}$. Since we are sampling points, $|L(\mathcal{H}_q)| = |L|$ and so (i) holds, and (v) follows from Lemma 1.

By the Chernoff bound (Proposition 1) with $\epsilon = 1/2$, the probability that $|P|$ is larger than $2aq^2 \log q$ or less than $(aq^2 \log q)/2$ is at most $2 \exp(-(aq^2 \log q)/16) < 1/3$, and the number of sampled points on a given line $\ell \in L(\mathcal{H}_q)$ is at most $(a \log q)/2$ with probability at most $\exp(-(a \log q)/8) = q^{-a/8} < q^{-8}$. Since $|L(\mathcal{H}_q)| \leq q^4 < q^8/3$, the probability that (iii) fails is less than $1/3$.

If the number of $(s+1)$ -fans on some pair of lines ℓ and ℓ' is more than k , then we claim some line had more than $2a \log q$ points sampled on it. This happens with probability at most $\exp(-(a \log q)/4) < q^{-8}$ by the Chernoff Bound (Proposition 1) with $\epsilon = 1$. Since the number of pairs of intersecting lines in $\mathcal{H}_q$ is at most $(q^3 + 1) \cdot \binom{q^2}{2} \leq q^8/3$, (iv) then fails with probability less than $1/3$. We then conclude (ii) – (iv) hold simultaneously with positive probability.

To prove the claim, fix distinct lines ℓ and ℓ' in $\mathcal{H}_q$. For each $(s+1)$ -fan in $\mathcal{H}$ containing ℓ and ℓ' , either $\{p\} = \ell \cap \ell'$ is the point of concurrency or some other point of $\ell \cup \ell' \setminus \{p\}$ is the point of concurrency. If p is the point of concurrency, then we must pick a point on ℓ and a point in ℓ' to define a line ℓ'' through the remaining points of the $(s+1)$ -fan. There are at most $(2a \log q)^2$ choices for those two points, and then since ℓ'' has at most $2a \log q$ points sampled, there are at most $(2a \log q)^{s-2}$ choices for the remaining $s-2$ points of the $(s+1)$ -fan. If p is not the point of concurrency, then the point of concurrency is picked from $\ell \cup \ell' \setminus \{p\}$ in at most $2(2a \log q)$ ways. Then we must pick the remaining $s-1$ points of the $(s+1)$ -fan from ℓ or from ℓ' , in at most $2(2a \log q)^{s-1}$ ways. So the number of $(s+1)$ -fans containing ℓ and ℓ' is at most $4(2a \log q)^s \leq (2a \log q)^s = k$. $\square$

For a point $p \in P$ and a set $X \subseteq L$, let X_p denote the set of lines X concurrent with the point p . It is convenient for a positive integer b and $X \subseteq L$ to define

$$P_X = P_{X,b} = \{p \in P : |X_p| \geq b\}.$$

Lemma 3. *Let $b \geq 1$, $a \geq 128$ and $q \geq a \log q$. Then for any $X \subseteq L$,*

$$\sum_{p \in P_X} |X_p| > \frac{1}{2}(a \log q) \cdot |X| - 2abq^2 \log q. \quad (6)$$

Proof. As $|P| \leq 2aq^2 \log q$ from Lemma 2.ii,

$$\sum_{p \in P \setminus P_X} |X_p| < b|P| \leq 2abq^2 \log q.$$

Each line in X is incident with at least $(a \log q)/2$ points by Lemma 2.iii. Therefore

$$\sum_{p \in P} |X_p| = \sum_{\ell \in X} |\ell \cap P| \geq \frac{1}{2}(a \log q) \cdot |X|.$$

Subtracting the first inequality from the second gives the lemma. $\square$

4.2 Random sampling of pairs of lines

We use the partial linear space $\mathcal{H} = \mathcal{H}_{a,q,s}$ in Lemma 2 for $a \geq 128$ and $q \geq a \log q$ to construct the *intersection graph* $G = G_{a,q,s}$ of lines in $\mathcal{H}$ as follows: the vertex set of G is L and the edge set is $\{\{\ell, \ell'\} : \ell \cap \ell' \neq \emptyset\}$. We use the words *line* and *vertex* interchangeably to refer to the vertices of G . From Lemma 2, for each prime power $q \geq a \log q$, the graph G has $|L| = q^2(q^2 - q + 1)$ vertices, and most importantly, for $s \geq 3$,

$$\begin{aligned} &\text{the vertex set of each } K_{s+1} \subseteq G \text{ is either an } (s+1)\text{-fan in } \mathcal{H} \text{ or} \\ &\text{is a set of lines of } \mathcal{H} \text{ all concurrent with some point in } \mathcal{H}. \end{aligned} \quad (\text{A})$$

For each point $p \in P$, the set of lines concurrent with p induces a clique of size q^2 in G , and any two of these cliques are edge-disjoint. Fixing a set $X \subseteq L$ and a point $p \in P$, recall $X_p \subseteq X$ is the set of lines in X concurrent with p , and in particular for $X_p \neq \emptyset$ the subgraphs $G[X_p]$ are edge-disjoint cliques for $p \in P$.

Then let $G_\chi \subseteq G$ be obtained from G by taking independently for each $p \in P$ a random vertex-coloring χ_p of all the lines in G concurrent with p and removing all edges of G whose ends have the same color. This removes from G all copies of K_{s+1} induced by $s+1$ lines concurrent with a single point. We conclude by (A), for $s \geq 3$:

$$\text{the vertex set of each } K_{s+1} \subseteq G_\chi \text{ is an } (s+1)\text{-fan}. \quad (\text{B})$$

We now define the random graph $G_\rho \subseteq G$. Let $b \geq 1$ and $\rho \in [0, 1]$ satisfy $b \geq 2^{40s}$ and $\rho = (8s/b)^{2/s}$, and define G_ρ to be the random graph obtained by sampling edges of G independently with probability ρ . We let H be the intersection of G_ρ and G_χ , namely, the graph with vertex set $V(G)$ and edge set $E(G_\rho) \cap E(G_\chi)$. Our next task is to consider copies of $K_s \subseteq H$ whose vertices are contained in some set of lines concurrent with a single point. Unsurprisingly, this makes strong use of Janson's inequality for the probability that a random s -partite graph is K_s -free, in the form of Proposition 2.

4.3 The event A_X

In this section, for each set X of lines, we define an event A_X whose non-occurrence implies $H[X]$ contains a copy of K_s , and we find an upper bound on $\Pr(A_X)$. For a set $X \subseteq L$ and a point $p \in P_X$, fix a family $\Pi_p(X) = \Pi_p$ of $r_p(X) = \lfloor |X_p|/b \rfloor$ disjoint subsets of X_p of size b each. Then X_p is *bad* if for $Y \in \Pi_p$, none of $H[Y]$ contains a K_s , and we let $A_{X,p}$ be the event that X_p is bad. We say X is *bad* if all $X_p : p \in P_X$ are bad. In other words, if A_X is the event that X is bad, then

$$A_X = \bigcap_{p \in P_X} A_{X,p}.$$

If A_X does not occur, then by definition $H[X]$ contains a copy of K_s .

Lemma 4. *Let $s \geq 3$, $b \geq 2^{40s}$ and $\rho \in [0, 1]$ satisfy $\rho = (8s/b)^{2/s}$. Then for any $X \subseteq L$,*

$$\Pr(A_X) \leq \exp\left(-\frac{1}{32s} \sum_{p \in P_X} |X_p|\right). \quad (7)$$

Proof. Since the colorings χ_p are independent over $p \in P_X$, the events $A_{X,p}$ are independent over $p \in P_X$. For $Y \in \Pi_p = \Pi_p(X)$, the events A_Y that $H[Y]$ does not contain K_s are independent. By the Chernoff bound (Proposition 1), the probability that χ_p assigns some color fewer than $b/2s$ times to vertices of Y is at most $\exp(-b/8s)$. Fix a coloring χ of Y where every color appears at least $b/2s$ times. Then the graph $H[Y]$ is a random s -partite graph which we denoted in Section 2 by $G_{b,\rho}(\chi)$. By Proposition 2, for any such coloring χ ,

$$\Pr(K_s \not\subseteq G_{b,\rho}(\chi)) \leq \exp(-2^{2s-4}b).$$

As there are at most s^b choices of χ , the union bound over s -colorings χ gives

$$\Pr(A_Y) \leq \exp\left(-\frac{b}{8s}\right) + s^b \exp(-2^{2s-4}b).$$

Since $s \geq 3$, $2^{2s-4}b \geq 2b \log s \geq 2b$ and therefore

$$\Pr(A_Y) \leq \exp\left(-\frac{b}{8s}\right) + \exp(-b) \leq 2 \exp\left(-\frac{b}{8s}\right) \leq \exp\left(-\frac{b}{16s}\right).$$

Here we used $b \geq 16s$. Since $|\Pi_p| = r_p(X)$, we obtain

$$\begin{aligned} \Pr(A_X) &= \prod_{p \in P_X} \Pr(A_{X,p}) \\ &= \prod_{p \in P_X} \prod_{Y \in \Pi_p} \Pr(A_Y) \leq \prod_{p \in P_X} \exp\left(-\frac{b}{16s} \cdot r_p(X)\right). \end{aligned}$$

Recall $|X_p| \geq b$ for $p \in P_X$, so $b \cdot r_p(X) \geq |X_p|/2$, and therefore

$$\Pr(A_X) \leq \prod_{p \in P_X} \exp\left(-\frac{|X_p|}{32s}\right) = \exp\left(-\frac{1}{32s} \sum_{p \in P_X} |X_p|\right).$$

This proves the lemma. □

5 Proof of Theorem 1

To prove Theorem 1, for each $s \geq 3$ let $G = G_{a,q,s}$ be the intersection graph defined in Section 4.2, where $a = 2^{10}s$ and q is a prime power satisfying $q \geq a \log q$. Let $H \subseteq G$ denote the random graph defined in Section 4.2, which is the intersection of the two random graphs G_χ and G_ρ , with parameters

$$b = 2^{40s} \cdot 2a \log q \quad \text{and} \quad \rho = \left(\frac{8s}{b}\right)^{\frac{2}{s}}.$$

For convenience we omit rounding and assume b is an integer. Let $\mathcal{K}$ be the family of sets of $s+1$ lines $K \subseteq L$ forming an $(s+1)$ -fan in G , so $G[K]$ is a complete graph of order $s+1$, and let A_K be the event $G[K] \subseteq G_\rho$. Let $\mathcal{X} = \{X \subseteq L : |X| = 8bq^2\}$ and A_X be the event X is bad, as in Section 4.3. Due to (B),

$$\begin{aligned} &\text{if none of the events } A_K : K \in \mathcal{K} \text{ or } A_X : X \in \mathcal{X} \text{ occur,} \\ &\text{then } H \text{ is } K_{s+1}\text{-free and } K_s \subseteq H[X] \text{ for all } X \in \mathcal{X}. \end{aligned} \tag{C}$$

Specifically, (C) implies every set of $8bq^2$ vertices of H induces a subgraph containing K_s . This shows for any prime power $q \geq a \log q$,

$$f_s(q^2(q^2 - q + 1)) \leq 8bq^2.$$

By Bertrand's postulate, there exists a prime between any positive integer and its double, so letting $q \geq a \log q$ be a prime between $n^{1/4}$ and $2n^{1/4}$, we find for $s \geq 3$ and $n \geq 2$,

$$\begin{aligned} f_s(n) &\leq 8bq^2 \leq 8 \cdot 2^{40s} \cdot 2^{11}s \cdot q^2 \cdot \log q \\ &\leq 2^{100s} \cdot \sqrt{n} \log n. \end{aligned}$$

It remains to prove (C) holds with positive probability, via the local lemma (Proposition 3).

Dependencies. For the dependencies between the events $A_K : K \in \mathcal{K}$ and $A_X : X \in \mathcal{X}$, we note A_X is determined by the following set of edges of G :

$$\hat{E}(X) = \bigcup_{p \in P_X} \bigcup_{Y \in \Pi_p} E(G[Y]).$$

Since $|Y| = b$ for each $Y \in \Pi_p$, and $r = r_p(X) = |\Pi_p| = \lfloor |X_p|/b \rfloor$,

$$\begin{aligned} |\hat{E}(X)| &= \sum_{p \in P_X} \sum_{i=1}^r \binom{b}{2} \\ &= \sum_{p \in P_X} \left\lfloor \frac{|X_p|}{b} \right\rfloor \binom{b}{2} \leq \frac{1}{2}b \sum_{p \in P_X} |X_p|. \end{aligned} \tag{8}$$

For convenience, let $\hat{E}(K)$ also denote $E(G[K])$ if $K \in \mathcal{K}$. It is important here that since G_ρ samples edges of G independently with probability ρ , for any $\mathcal{J} \subseteq \mathcal{K}$ and $K \in \mathcal{K}$, we observe

the event A_K is mutually independent with $\{A_{K'} : K' \in \mathcal{J}\}$ if

$$\bigcup_{K' \in \mathcal{J}} E(K') \cap E(K) = \emptyset.$$

Let $k = (2a \log q)^s$. By Lemma 2.iv, each edge of G is contained in at most k cliques $K \in \mathcal{K}$. Fixing any $K \in \mathcal{K}$, there are at most

$$\kappa = \binom{s+1}{2} \cdot k \leq bk \quad (9)$$

choices of $K' \in \mathcal{K}$ such that $\hat{E}(K') \cap \hat{E}(K) \neq \emptyset$, and any other set of events $A_{K'}$ are mutually independent with A_K by the observation above. We assume all A_X are mutually dependent with any given event A_K .

Since each edge of G is in at most k cliques in $\mathcal{K}$, by (8), for each $X \in \mathcal{X}$ there are at most

$$\begin{aligned} \lambda &= k \cdot |\hat{E}(X)| \\ &\leq \frac{1}{2}bk \cdot \sum_{p \in P_X} |X_p| \end{aligned} \quad (10)$$

choices of $K \in \mathcal{K}$ such that $\hat{E}(K) \cap \hat{E}(X) \neq \emptyset$, and any set of other events A_K are mutually independent with A_X . We assume all $A_{X'}$ are mutually dependent with any given event A_X .

Local lemma inequalities. Let $N = |\mathcal{X}|$. The local lemma (Proposition 3) implies the probability that (C) holds is positive if (5) holds, i.e. there are reals $\gamma, \delta \in [0, 1)$ such that for all $K \in \mathcal{K}$ and all $X \in \mathcal{X}$,

$$\Pr(A_K) \leq \gamma(1 - \gamma)^\kappa(1 - \delta)^N \quad \text{and} \quad \Pr(A_X) \leq \delta(1 - \gamma)^\lambda(1 - \delta)^N.$$

We claim that these inequalities are satisfied if we select $\delta = 1/(N + 1)$ and $\gamma = 1/32sbk$.

First inequality. We have $\Pr(A_K) = \rho^{\binom{s+1}{2}}$ and $(1 - \delta)^N \geq 1/e$. By (9), $(1 - \gamma)^\kappa \geq 1 - \kappa\gamma > 1/2$, so it is sufficient to show $2e \Pr(A_K)/\gamma \leq 1$ for the first inequality to hold. Since $\rho = (8s/b)^{2/s}$,

$$\rho^{\binom{s+1}{2}} = \left(\frac{8s}{b}\right)^{s+1}.$$

Since $b = 2^{40s} \cdot 2a \log q = 2^{40s} k^{1/s}$, and $s \geq 3$,

$$\begin{aligned} \frac{2e}{\gamma} \Pr(A_K) &= 64esbk \cdot \rho^{\binom{s+1}{2}} \\ &\leq \frac{64esk(8s)^{s+1}}{b^s} \\ &= \frac{64esk(8s)^{s+1}}{2^{40s^2}k} \\ &< \frac{64esk(8s)^{s+1}}{64^{s^2}k} \\ &= es \cdot \left(\frac{8s}{64^{s-1}}\right)^{s+1} \\ &< \frac{es}{8^{s+1}} < 1. \end{aligned}$$

Here we used $64^{s-1} > 64s$ and $es < 8^{s+1}$ for $s \geq 3$. This verifies the first inequality.

Second inequality. For the second inequality, we use $1 - \gamma \geq \exp(-2\gamma)$, which is valid since $\gamma \leq 1/2$. Recalling $(1 - \delta)^N \geq 1/e$, it is enough to show

$$e \cdot \Pr(A_X) \leq \exp(-\log(N + 1) - 2\gamma\lambda).$$

By Lemma 4,

$$\Pr(A_X) \leq \exp\left(-\frac{1}{32s} \sum_{p \in P_X} |X_p|\right).$$

Using $|L| = |V(G)| = q^2(q^2 + q + 1)$,

$$\log(N + 1) = \log\left[\left(\frac{q^2(q^2 - q + 1)}{8bq^2}\right) + 1\right] \leq \log\left(\frac{q^4}{8bq^2}\right) - 1 \leq 32bq^2 \log q - 1.$$

Due to the upper bound on λ given by (10), it is enough to show

$$\exp\left(-\frac{1}{32s} \sum_{p \in P_X} |X_p|\right) \leq \exp\left(-32bq^2 \log q + \frac{1}{64s} \sum_{p \in P_X} |X_p|\right).$$

Therefore we require

$$\exp\left(-\frac{1}{64s} \sum_{p \in P_X} |X_p|\right) \leq \exp(-32bq^2 \log q).$$

Applying (6) and using $a = 2^{10}s$, we find

$$\begin{aligned} \frac{1}{64s} \sum_{p \in P_X} |X_p| &\geq \frac{1}{64s} \left(\frac{1}{2} (a \log q) \cdot |X| - 2abq^2 \log q \right) \\ &= \frac{1}{64s} \left(\frac{1}{2} (a \log q) \cdot 8bq^2 - 2abq^2 \log q \right) \\ &= \frac{1}{64s} \cdot 2abq^2 \log q = 32bq^2 \log q. \end{aligned}$$

This proves the second inequality. □

Concluding remarks

- In this paper, we proved $f_s(n) = O(\sqrt{n} \log n)$ by suitable random sampling of points and lines from Hermitian unitals. A part of the proof essentially involves the union bound over all sets X of lines of size $8bq^2$ – we implicitly assumed in our application of the local lemma in Section 5 that the events A_X depend on all other such events. We first sampled points randomly from the Hermitian unital with probability of order $(\log q)/q$. If we sampled points with a lower probability $o(\log q)/q$, then the union bound no longer works: for large q there are $N \geq \exp(bq^2 \log q)$ sets of size $8bq^2$ to consider – see Section 5 – whereas if all the sets $X_p : p \in P_X$ have size roughly b , then the probability that every X_p is $(s - 1)$ -colored in a random s -coloring is $\exp(-o(bq^2 \log q))$. We believe it may be possible using randomized greedy algorithms akin to the Rödl semirandom method or more recent iterative absorption methods to circumvent this issue and obtain a bounds $f_s(n) = o(\sqrt{n} \log n)$, but did not investigate this technical direction.

• It is plausible that when $s \geq 3$, $f_s(n) = \Omega(\sqrt{n}(\log n)^\alpha)$ for some $\alpha > 1/2$. The current lower bound of order $n\sqrt{\log n}/\log \log n$ is achieved by finding an induced K_s -free subgraph of size d (the neighborhood of a vertex of degree d) or an independent set of size $n(\log d)/d(\log \log d)$ in a K_{s+1} -free graph with maximum degree d . The latter seems potentially wasteful, in that we could perhaps find for $s \geq 3$ an induced K_s -free subgraph of size much larger than $n(\log d)/d(\log \log d)$ in a K_{s+1} -free graph. In addition, it is believable that the $\log \log d$ term is superfluous; perhaps every K_{s+1} -free graph with maximum degree d has an independent set of order at least $n(\log d)/d$.

6 Appendix : Proof of Proposition 2

To prove Proposition 2, we require some notation and preliminaries. Recall for $s \geq 3$, $n \geq 1$, $\rho \in [0, 1]$ and an s -coloring χ of an n -element set with color classes $Y_1, Y_2, \dots, Y_s$, $G_{n,\rho}(\chi)$ is obtained by independently and randomly sampling edges of the complete s -partite graph with parts $Y_1, Y_2, \dots, Y_s$ with probability ρ . The expected number of $K_s \subseteq G_{n,\rho}(\chi)$ is precisely

$$\mu(\chi) = \rho^{\binom{s}{2}} \prod_{i=1}^s |Y_i| \quad (11)$$

and the variance is defined by

$$\Delta(\chi) = \sum_{S \subset [s]} \rho^{2\binom{s}{2} - \binom{|S|}{2}} \prod_{i \in S} |Y_i| \prod_{j \notin S} |Y_j| (|Y_j| - 1) = \sum_{S \subset [s]} \rho^{2\binom{s}{2} - \binom{|S|}{2}} \prod_{i=1}^n |Y_i| \prod_{j \notin S} (|Y_j| - 1), \quad (12)$$

where the sum is over sets S with $2 \leq |S| \leq s-1$. From Janson's inequality [2, 9] one obtains for any $\rho \in [0, 1]$ and $n \geq 1$:

$$\Pr(K_s \not\subseteq G_{n,\rho}(\chi)) \leq \exp\left(-\frac{1}{2}\mu(\chi)\right). \quad (13)$$

provided $\Delta(\chi) \leq \mu(\chi)$.

Proof of Proposition 2. For Proposition 2, χ is an s -coloring with color classes $Y_1, Y_2, \dots, Y_s$ satisfying $|Y_i| \geq n/2s$ for all $i \in [s]$. The product in (11) is minimized when $|Y_i| = n/2s$ for all but one value of $i \in [s]$, and the remaining color class has size $n - (s-1)n/2s > n/2$. Therefore

$$\begin{aligned} \mu(\chi) &> \rho^{\binom{s}{2}} \left(\frac{n}{2s}\right)^{s-1} \cdot \frac{n}{2} \\ &= \left(\frac{8s}{n}\right)^{s-1} \left(\frac{n}{2s}\right)^{s-1} \cdot \frac{n}{2} = 2^{2s-3} \cdot n. \end{aligned}$$

So if $\Delta(\chi) \leq \mu(\chi)$ when $\rho = (8s/n)^{2/s}$ and $n \geq 2^{40s}$, then Proposition 2 follows from (13):

$$\Pr(K_s \not\subseteq G_{n,\rho}(\chi)) \leq \exp\left(-\frac{1}{2}\mu(\chi)\right) \leq \exp(-2^{2s-4}n).$$

It remains to prove $\Delta(\chi) \leq \mu(\chi)$ when $\rho = (8s/n)^{2/s}$ and $n \geq 2^{40s}$.

By (11) and (12),

$$\frac{\Delta(\chi)}{\mu(\chi)} = \sum_{S \subset [s]} \rho^{\binom{s}{2} - \binom{|S|}{2}} \prod_{j \notin S} (|Y_j| - 1).$$

The sum is over subsets S of $[s]$ where $2 \leq |S| \leq s-1$. By the inequality of geometric and arithmetic means,

$$\prod_{j \notin S} (|Y_j| - 1) \leq \prod_{j \notin S} |Y_j| \leq \left(\frac{1}{s-|S|} \sum_{j \notin S} |Y_j| \right)^{s-|S|} \leq \left(\frac{n}{s-|S|} \right)^{s-|S|}.$$

We conclude

$$\frac{\Delta(\chi)}{\mu(\chi)} \leq \sum_{i=2}^{s-1} \rho_{\binom{s}{2}-\binom{i}{2}} \left(\frac{n}{s-i} \right)^{s-i} \binom{s}{i}.$$

By definition of ρ ,

$$\rho_{\binom{s}{2}-\binom{i}{2}} = \left(\frac{8s}{n} \right)^{\frac{(s-i)(s+i-1)}{s}}.$$

Therefore

$$\frac{\Delta(\chi)}{\mu(\chi)} \leq \sum_{i=2}^{s-1} \left(\frac{8s}{n} \right)^{\frac{(s-i)(s+i-1)}{s}} \left(\frac{n}{s-i} \right)^{s-i} \binom{s}{i} = \sum_{i=2}^{s-1} \left(\frac{(8s)^{\frac{s+i-1}{s}}}{(s-i)n^{\frac{i-1}{s}}} \right)^{s-i} \binom{s}{i}.$$

We break the sum into two pieces. First, for $2 \leq i \leq \lfloor s/\log(8s) \rfloor \leq s/2$,

$$(s-i)n^{\frac{i-1}{s}} \geq \frac{s}{2} \cdot n^{\frac{1}{s}} \geq 2^9 s$$

since $n \geq 2^{40s} \geq 2^{10s}$. Therefore each term in the sum is at most

$$\left(\frac{8s \cdot (8s)^{\frac{1}{\log(8s)}}}{2^9 s} \right)^{s-i} \binom{s}{i} \leq \left(\frac{8es}{2^9 s} \right)^{s-\frac{s}{\log(8s)}} \cdot 2^s \leq \left(\frac{1}{16} \right)^{\frac{s}{2}} \cdot 2^s \leq 2^{-s}.$$

Second, for $\lfloor s/\log(8s) \rfloor + 1 < i \leq s-1$, $(i-1)/s \geq 1/2 \log(8s)$ and so using $n \geq 2^{40s}$ and $s \geq 3$,

$$(s-i)n^{\frac{i-1}{s}} \geq n^{\frac{1}{2 \log(8s)}} \geq 2^{\frac{20s}{\log(8s)}} \geq 128s^3.$$

Therefore each term in the sum is at most

$$\left(\frac{(8s)^{\frac{s+i-1}{s}}}{128s^3} \right)^{s-i} \binom{s}{i} \leq \left(\frac{(8s)^2}{128s^3} \right)^{s-i} \binom{s}{i} = \left(\frac{1}{2s} \right)^{s-i} \binom{s}{i}.$$

We conclude

$$\begin{aligned} \frac{\Delta(\chi)}{\mu(\chi)} &\leq \sum_{i=2}^{s-1} \left(\frac{1}{2s} \right)^{s-i} \binom{s}{i} + \sum_{i=2}^{s-1} 2^{-s} \\ &\leq \left(1 + \frac{1}{2s} \right)^s - 1 + (s-2)2^{-s} \leq \sqrt{e} - 1 + (s-2)2^{-s}. \end{aligned}$$

Evidently $(s-2)2^{-s} \leq 1/8$ and therefore

$$\sqrt{e} - 1 + (s-2)2^{-s} \leq 0.773 \dots < 1.$$

We conclude $\Delta(\chi) < \mu(\chi)$, as required. $\square$

References

- [1] M. Ajtai, J. Komlós, and E. Szemerédi, A note on Ramsey numbers, *J. Combin. Theory Ser. A* 29 (1980), no. 3, 354–360.
- [2] N. Alon and J. Spencer, *The probabilistic method*, (Third Edition) John Wiley and Sons, New York, 2008.
- [3] S. G. Barwick, G. L. Ebert, *Unitals in projective planes*, Springer Monographs in Mathematics. Springer, New York, 2008. xii+193 pp.
- [4] T. Bohman and P. Keevash, The early evolution of the H -free process, *Invent. Math.*, 181 (2010), 291–336.
- [5] A. E. Brouwer, and H. Van Maldeghem, *Strongly Regular Graphs*, Cambridge Univ. Press, 2022.
- [6] A. Dudek, T. Retter, and V. Rödl, On generalized Ramsey numbers of Erdős and Rogers, *J. Combin. Theory Ser. B* 109 (2014), 213–227.
- [7] G. Fiz Pontiveros, S. Griffiths and R. Morris The triangle-free process and the Ramsey numbers, *Mem. Amer. Math. Soc.*, 263 (2020), 125pp.
- [8] O. Janzer, T. Gowers, Improved bounds for the Erdős-Rogers function, *Advances in Combinatorics*, 2020 (3), 27 pp.
- [9] S. Janson, T. Łuczak, and A. Ruciński, *Random graphs*, Wiley (2000).
- [10] O. Janzer, B. Sudakov, Improved bounds for the Erdős-Rogers $(s, s + 2)$ -problem, Preprint at <https://arxiv.org/abs/2307.05441>.
- [11] J. H. Kim, The Ramsey number $R(3, t)$ has order of magnitude $t^2/\log t$, *Random Struct. Alg.* 7 (1995), 173–207.
- [12] M. Krivelevich, Bounding Ramsey numbers through large deviation inequalities, *Random Struct. Alg.* 7 (1995), no.2, 145–155.
- [13] P. Erdős, L. Lovász, Problems and results on 3-chromatic hypergraphs and some related questions, In A. Hajnal; R. Rado; V. T. Sós (eds.). *Infinite and Finite Sets* (to Paul Erdős on his 60th birthday). Vol. II. North-Holland. pp. 609–627.
- [14] S. Mattheus, and J. A. Verstraëte, The asymptotics of $r(4, t)$, to appear, *Annals of Math.*
- [15] D. Mubayi, and J. A. Verstraete, A note on pseudorandom Ramsey graphs, to appear, *J. Eur. Math. Soc. (JEMS)* (10 pages).
- [16] M. E. O’Nan, Automorphisms of unitary block designs, *J. Algebra* 20 (1972), 495–511.
- [17] F. C. Piper, Unitary block designs, pp. 98–105 in: *Graph Theory and Combin.* (Proc. Milton Keynes, 1978), R. J. Wilson (ed.), Res. Notes in Math. 34, Pitman, Boston, 1979. (p. 85)

- [18] J. B. Shearer, A note on the independence number of triangle-free graphs, *Discrete Math.*, 46 (1983), 83–87.
- [19] J. B. Shearer, On the independence number of sparse graphs, *Random Struct. Alg.* 7 (1995), no. 3, 269–271.
- [20] M. Talagrand, Concentration of measure and isoperimetric inequalities in product spaces, *Publications Mathématiques de l’IHÉS. Springer-Verlag* (1995) 81, 73–205.
- [21] G. Wolfowitz, K_4 -free graphs without large induced triangle-free subgraphs, *Combinatorica* 33 (2013), no. 5, 623–631.